Lecture Scribe: Joint Random Variables and Distributions

Scribe: Vansh Lilani (ID: AU2320146)

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Lecturer: Dhaval Patel, PhD

1 Introduction to Joint Random Variables

In scientific analysis and real-world applications, it is often necessary to observe and analyze two or more random variables (RVs) simultaneously to understand their relationships.

Real-World Examples

- **Smart Transportation:** X = Vehicle Speed, Y = Traffic Density.
- **Wireless Networks:** X = Signal-to-Noise Ratio (SNR), Y = Data Throughput.
- **Logistics:** X = Wind Speed, Y = Delivery Time for drone delivery.
- **General Biology/Social Science:** Height and weight of giraffes, IQ and birthweight of children, or exercise frequency and heart disease rates.

Key Concepts

- **Joint Distribution:** Allows the computation of probabilities for events involving multiple variables and reveals their dependency.
- **Independence:** The simplest case where variables do not influence each other.
- **Correlation/Covariance:** Measures used to quantify the nature of dependence when variables are not independent.

2 Discrete Case: Joint Probability Mass Function (PMF)

Setup and Definition

Suppose X and Y are discrete random variables with possible values $X \in \{x_1, x_2, \ldots, x_n\}$ and $Y \in \{y_1, y_2, \ldots, y_m\}$. The ordered pair (X, Y) takes values in the product set $\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$.

Definition: The Joint PMF is defined as:

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This represents the probability that the joint outcome occurs simultaneously.

Key Properties

1. **Non-negativity:** $0 \le p(x_i, y_j) \le 1$ for all i, j.
2. **Normalization:** The sum of all joint probabilities must equal 1:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

3 Worked Example: Rolling Two Dice

Experiment: Roll two fair dice. Let X = Value on the first die and T = Total (sum) of both dice.
Construction: Since each outcome (i, j) of two dice has a probability of 1/36, we construct the joint probability table for (X, T).

$X \setminus T$	2	3	4	5	6	7	8	9	10	11	12
1	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0	0	0
2	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0	0
3	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0
4	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0
5	0	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0
6	0	0	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36

Observations:

- Entries are either 0 or 1/36.
- X and T are **NOT** independent because the possible values of T depend on the value of X.

4 Continuous Case: Joint Probability Density Function (PDF)

The continuous case transitions from sums to integrals:

- Discrete values $\to$ Continuous intervals.
- Joint PMF $\to$ Joint PDF.
- Sums $\to$ Double integrals.

Setup and Definition

If $X \in [a, b]$ and $Y \in [c, d]$, the pair (X, Y) takes values in the product region $[a, b] \times [c, d]$.
Definition: A function $f(x, y)$ is the joint PDF of X and Y if:

$$P((X, Y) \in \text{small rectangle area } dx\, dy) \approx f(x, y)\, dx\, dy$$

Key Properties

1. **Non-negativity:** $f(x, y) \ge 0$.
2. **Normalization:** $\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$.
3. **Density vs. Probability:** $f(x, y)$ can be > 1 as it is a density.

5 Comprehensive Example: Continuous Joint PDF

Given: $f(x, y) = 4xy$ for $0 \leq x \leq 1$ and $0 \leq y \leq 1$.

1. Verification of Validity

Check if the double integral over the unit square $[0, 1]^2$ equals 1:

$$\int_0^1 \int_0^1 4xy\,dx\,dy = \int_0^1 \left[2x^2 y\right]_{x=0}^{x=1} dy = \int_0^1 2y\,dy = \left[y^2\right]_0^1 = 1$$

2. Computing Probability $P(A)$

For $A = \{X < 0.5, Y > 0.5\}$:

$$P(A) = \int_{0.5}^1 \int_0^{0.5} 4xy\,dx\,dy$$

Inner integral (x): $\int_0^{0.5} 4xy\,dx = [2x^2 y]_0^{0.5} = 2(0.25)y = 0.5y$
Outer integral (y): $\int_{0.5}^1 0.5y\,dy = [0.25y^2]0.5^1 = 0.25(1 - 0.25) = 0.1875$.

6 Joint Cumulative Distribution Function (Joint CDF)

Definition: $F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$

Calculation Methods

- **Continuous:** $F_{X,Y}(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(u, v)\,du\,dv$
- **Discrete:** $F_{X,Y}(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$

Properties of Joint CDF

- **Monotonicity:** $F(x, y)$ is non-decreasing in both x and y.
- **Boundary:** $F(-\infty, y) = 0$, $F(x, -\infty) = 0$, and $F(\infty, \infty) = 1$.
- **Relationship to PDF:** $f(x, y) = \frac{\partial^2 F(x,y)}{\partial x \partial y}$.